In addition, explain with distributions are assumed

Hypothesis testing methods rely on specific assumptions about the underlying distributions of data. These assumptions are critical for the validity of the test results. Below is an explanation of the distributions assumed by common hypothesis testing methods, categorized by parametric and non-parametric approaches.

1. Parametric Tests and Their Assumptions

Parametric tests assume that the data follow a specific probability distribution, most commonly the normal distribution. These tests are more powerful when their assumptions are met but may produce invalid results if the assumptions are violated.

Z-Test

- Assumptions:
  - The population from which the sample is drawn is normally distributed.
  - The population standard deviation (σ) is known.
  - For large sample sizes ($n \geq 30$), the Central Limit Theorem ensures that the sampling distribution of the mean approximates normality, even if the population distribution is not normal [1] [2] .
- Distribution: Under the null hypothesis (H_0), the test statistic follows a standard normal distribution ($N(0, 1)$) [1] [7] .

T-Test

- Assumptions:
  - The population from which the sample is drawn is normally distributed.
  - The population standard deviation is unknown, and the sample standard deviation (s) is used as an estimate.
  - Data are randomly sampled and independent.
- Distribution: The test statistic follows a Student's t-distribution with $n - 1$ degrees of freedom. For large sample sizes, this approximates the standard normal distribution [1] [4] [7] .

ANOVA (Analysis of Variance)

- Assumptions:
  - Each group being compared comes from a normally distributed population.
  - Homogeneity of variance: All groups have equal variances.
  - Observations are independent.
- Distribution: The test statistic follows an F-distribution under H_0 [4] .

Chi-Square Test

- Assumptions:
  - Data are categorical, and observations are independent.
  - Expected frequencies in each category should be sufficiently large (typically at least 5).
- Distribution: The test statistic follows a chi-square (χ^2) distribution under H_0 [1] [4] .

2. Non-Parametric Tests and Their Assumptions

Non-parametric tests do not assume specific distributions for the data, making them more robust to violations of normality or other parametric assumptions. These tests are often referred to as "distribution-free."

Mann-Whitney U Test

- Assumptions:
  - Samples are independent and come from populations with similar shapes.
  - Data can be ranked (ordinal scale).
- Distribution: No specific distribution is assumed; it evaluates ranks rather than raw data values [3] [6] .

Kruskal-Wallis Test

- Assumptions:
  - Samples are independent and come from populations with similar shapes and spreads.
  - Data can be ranked.
- Distribution: Like Mann-Whitney U, this test does not assume any specific distribution but uses ranks for analysis [3] [6] .

Wilcoxon Signed-Rank Test

- Assumptions:
  - Data come from paired samples or matched observations.
  - Differences between pairs can be ranked.
- Distribution: No specific distribution is assumed; it analyzes ranks of differences [3] [6] .

3. Key Considerations for Distribution Assumptions

Normality

Many parametric tests assume normality because their test statistics rely on properties of the normal distribution. If this assumption is violated, non-parametric alternatives like Mann-Whitney U or Kruskal-Wallis should be used [3] [9] .

Central Limit Theorem

For large sample sizes, parametric tests like z
-tests or t
-tests become robust to deviations from normality because the sampling distribution of the mean approaches normality regardless of the population distribution [1] [7] .

Homogeneity of Variance

Tests like ANOVA assume equal variances across groups. This can be checked using tests like Levene's test before applying ANOVA [4] .

Conclusion

The choice of hypothesis testing method depends on whether parametric assumptions (normality, known variance, etc.) hold true for the data. Parametric tests like z
-tests, t
-tests, and ANOVA require specific distributions (e.g., normal), while non-parametric tests like Mann-Whitney U or Kruskal-Wallis

do not assume any particular distribution, making them more flexible but less powerful when parametric assumptions are met.